Date:22/10/2026

10.Feature Engineering

Procedure:

Step 1: Import the required libraries, Pandas and NumPy.
Step 2: Read the training dataset using the pd.read_csv() function.
Step 3: Display the dataset information using dt.info() and check missing values using dt.isnull().sum().
Step 4: Define a function to handle missing values in continuous data and calculate the median of the Age column.
Step 5: Replace the missing values in the Age column with its median value.
Step 6: Define a function to replace missing categorical values with the most frequent category (mode).
Step 7: Apply the function to the Embarked, Cabin_Serial and Cabin columns to handle missing values.
Step 8: Create dummy variables for categorical data using pd.get_dummies().
Step 9: Perform ordinal encoding on the Pclass column by assigning First = 1, Second = 2 and Third = 3.
Step 10: Create a new column named Ord_Pclass, map the class categories to their corresponding ranks and drop the original Pclass column.
Step 11: Create a copy of the dataset named dt2 and identify the categorical columns.
Step 12: Perform count/frequency encoding by calculating the frequency of each category in the Salutation, Sex, Embarked, Cabin_Serial and Cabin columns.
Step 13: Create new columns by mapping the category frequencies to their respective columns and drop the original categorical columns.
Step 14: Create a copy of the dataset named dt3 and extract the first character of the Cabin column.
Step 15: Calculate the mean survival rate for each Cabin category using the Survived column.
Step 16: Sort the Cabin categories according to their mean survival rates and assign integer ranks to the categories.
Step 17: Create the Cabin_Encoded column by mapping the Cabin categories to their respective ranks and drop the original Cabin column.
Step 18: Create a copy of the dataset named dt4 and calculate the mean survival rate for each Cabin category.
Step 19: Perform mean encoding by mapping the mean survival rates to a new column named Cabin_Mean_Encoded and drop the original Cabin column.
Step 20: Create a copy of the dataset named dt5 and extract the first character of the Cabin column.
Step 21: Calculate the probability of survival for each Cabin category and store the results in a DataFrame named Probability_Survived.
Step 22: Calculate the probability of death by subtracting the survival probability from 1.
Step 23: Calculate the probability ratio by dividing the survival probability by the death probability for each Cabin category.
Step 24: Create a dictionary containing the probability ratios and map them to a new column named Encode_Prob_Ratio.
Step 25: Drop the original Cabin column and display the final encoded dataset.

Solution:

```
import pandas as pd
import numpy as np
file_path = r"C:\Users\Lenovo\Downloads\KAVYA SDM College(Autonomous), Ujire - train_FE - KAVYA SDM College(Autonomous), Ujire - train_FE.csv"
dt = pd.read_csv(file_path)
dt.info()
```

```
print("Initial Null Values:")
print(dt.isnull().sum())

# --- 1. Handle continuous missing data ---
def impute_nan(DataFrame, ColumnName, ImputeValue):
    DataFrame[ColumnName] = DataFrame[ColumnName].fillna(ImputeValue)
median = dt.Age.median()
impute_nan(dt, 'Age', median)
print("\nNull values after Age imputation:")
print(dt.isnull().sum())

# --- 2. Handle categorical missing data ---
def impute_nan_most_frequent_category(DF, ColName):
    most_frequent_category = DF[ColName].mode()[0]
    DF[ColName] = DF[ColName].fillna(most_frequent_category)
for Columns in ['Embarked', 'Cabin_Serial', 'Cabin']:
    impute_nan_most_frequent_category(dt, Columns)
print("\nNull values after categorical imputation:")
print(dt.isnull().sum())
dt_dummies = pd.get_dummies(dt)
print("\nOne-Hot Encoded Dummies Head:")
print(dt_dummies.head())

# --- 3. Ordinal Number Encoding ---
PClassDict = {'First': 1, 'Second': 2, 'Third': 3}
dt['Ord_Pclass'] = dt.Pclass.map(PClassDict)
print("\nUnique values in Ord_Pclass:", dt['Ord_Pclass'].unique())
print("\nOrdinal Encoding Result:")
print(dt[['Pclass', 'Ord_Pclass']].head(10))
dt = dt.drop('Pclass', axis=1)
print(dt.head(10))

# --- 4. Count / Frequency Encoding ---
dt2 = dt.copy()
print("\ndt2 Shape:", dt2.shape)
dt_cat = dt.select_dtypes(include=['object', 'category']).columns.tolist()
print("\nCategorical Columns:", dt_cat)
Salutation_Dict = dt2['Salutation'].value_counts()
Sex_Dict = dt2['Sex'].value_counts()
Embarked_Dict = dt2['Embarked'].value_counts()
Cabin_Serial_Dict = dt2['Cabin_Serial'].value_counts()
Cabin_Dict = dt2['Cabin'].value_counts()
print("\nSalutation Frequencies:")
print(Salutation_Dict)
dt2['Salutation_Dict'] = dt2['Salutation'].map(Salutation_Dict)
dt2['Sex_Dict'] = dt2['Sex'].map(Sex_Dict)
dt2['Embarked_Dict'] = dt2['Embarked'].map(Embarked_Dict)
dt2['Cabin_Serial_Dict'] = dt2['Cabin_Serial'].map(Cabin_Serial_Dict)
```

```
dt2['Cabin_Dict'] = dt2['Cabin'].map(Cabin_Dict)
print("\nFrequency Encoded Columns:")
print(dt2[['Salutation', 'Salutation_Dict', 'Sex', 'Sex_Dict', 'Embarked',
'Embarked_Dict',
           'Cabin_Serial', 'Cabin_Serial_Dict', 'Cabin', 'Cabin_Dict']].head())
dt2 = dt2.drop(['Salutation', 'Sex', 'Embarked', 'Cabin_Serial', 'Cabin'], axis=1)
print(dt2.head())

# --- 5. Target / Guided Encoding ---
dt3 = dt.copy()
dt3['Cabin'] = dt3['Cabin'].astype(str).str[0]
print("\nMean Survived per Cabin:")
print(dt3.groupby(['Cabin'])['Survived'].mean())
print("Overall Mean Survived:", dt3['Survived'].mean())
Encoded_Lables = dt3.groupby(['Cabin'])['Survived'].mean().sort_values().index
Encoded_Lables_Ranks = {k: i for i, k in enumerate(Encoded_Lables, 0)}
print("\nEncoded Labels Ranks:", Encoded_Lables_Ranks)
dt3['Cabin_Encoded'] = dt3['Cabin'].map(Encoded_Lables_Ranks)
print("\nTarget Encoding Result:")
print(dt3[['Cabin', 'Cabin_Encoded']].head(10))
dt3 = dt3.drop('Cabin', axis=1)

# --- 6. Mean Encoding ---
dt4 = dt.copy()
Encoded_Mean_Dict = dt4.groupby(['Cabin'])['Survived'].mean().to_dict()
dt4['Cabin_Mean_Encoded'] = dt4['Cabin'].map(Encoded_Mean_Dict)
print("\nMean Encoding Result:")
print(dt4[['Cabin', 'Cabin_Mean_Encoded']].head(10))
dt4 = dt4.drop('Cabin', axis=1)

# --- 7. Probability Ratio Encoding ---
dt5 = dt.copy()
dt5['Cabin'] = dt5['Cabin'].astype(str).str[0]
Probability_Survived = dt5.groupby(['Cabin'])['Survived'].mean()
Probability_Survived = pd.DataFrame(Probability_Survived)
Probability_Survived['Died'] = 1 - Probability_Survived['Survived']
Probability_Survived['Prob_Ratio'] = Probability_Survived['Survived'] /
Probability_Survived['Died']
print("\nProbability Survived / Died Dataframe:")
print(Probability_Survived)
Encode_Prob_Ratio = Probability_Survived['Prob_Ratio'].to_dict()
print("\nProbability Ratio Dictionary:", Encode_Prob_Ratio)
dt5['Encode_Prob_Ratio'] = dt5['Cabin'].map(Encode_Prob_Ratio)
print("\nProbability Ratio Encoding Result:")
print(dt5[['Cabin', 'Encode_Prob_Ratio']].head(10))
dt5 = dt5.drop('Cabin', axis=1)
print(dt5.head(10))
Data columns (total 14 columns):
```

```
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    str
 3   Salutation    891 non-null    str
 4   Name          891 non-null    str
 5   Sex           891 non-null    str
 6   Age           714 non-null    float64
 7   SibSp         891 non-null    int64
 8   Parch         891 non-null    int64
 9   Ticket        891 non-null    str
 10  Fare          891 non-null    float64
 11  Embarked      889 non-null    str
 12  Cabin_Serial  204 non-null    str
 13  Cabin         204 non-null    str
dtypes: float64(2), int64(4), str(8)
memory usage: 97.6 KB
Initial Null Values:
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age             177
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
Null values after Age imputation:
```

```
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age               0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
Null values after categorical imputation:
PassengerId     0
Survived        0
Pclass          0
Salutation      0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
Cabin_Serial    0
Cabin           0
dtype: int64
One-Hot Encoded Dummies Head:
```

PassengerId	Survived	Age	SibSp	Parch	Fare	Pclass_First	Pclass_Second	...	Cabin_F G63	Cabin_F G73	Cabin_F2	Cabin_F33	Cabin_F38	Cabin_F4	Cabin_G6	Cabin_T
0	1	0	22.0	1	0	7.2500	False	False	...	False	False	False	False	False	False	False	False

1 2 1 38.0 1 0 71.2833 True False ... False False False False False False False False

2 3 1 26.0 0 0 7.9250 False False ... False False False False False False False False

3 4 1 35.0 1 0 53.1000 True False ... False False False False False False False False

4 5 0 35.0 0 0 8.0500 False False ... False False False False False False False False

```
Unique values in Ord_Pclass: [3 1 2]

Ordinal Encoding Result:

   Pclass  Ord_Pclass

0   Third           3

1   First           1

2   Third           3

3   First           1

4   Third           3

5   Third           3

6   First           1

7   Third           3

8   Third           3

9  Second           2
```

	PassengerId	Survived	Salutation	Name	Sex	Age	...	Ticket	Fare	Embarked	Cabin_Serial	Cabin	Ord_Pclass
0	1	0	Mr	Braund, Mr. Owen Harris	male	22.0	...	A/5 21171	7.2500	S	C	B96 B98	3
1	2	1	Mrs	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	...	PC 17599	71.2833	C	C	C85	1
2	3	1	Miss	Heikkinen, Miss. Laina	female	26.0	...	STON/O2. 3101282	7.9250	S	C	B96 B98	3
3	4	1	Mrs	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	...	113803	53.1000	S	C	C123	1
4	5	0	Mr	Allen, Mr. William Henry	male	35.0	...	373450	8.0500	S	C	B96 B98	3
5	6	0	Mr	Moran, Mr. James	male	28.0	...	330877	8.4583	Q	C	B96 B98	3
6	7	0	Mr	McCarthy, Mr. Timothy J	male	54.0	...	17463	51.8625	S	E	E46	1
7	8	0	Master	Palsson, Master. Gosta Leonard	male	2.0	...	349909	21.0750	S	C	B96 B98	3
8	9	1	Mrs	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	...	347742	11.1333	S	C	B96 B98	3
9	10	1	Mrs	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	...	237736	30.0708	C	C	B96 B98	2

dt2 Shape: (891, 14)

Categorical Columns: ['Salutation', 'Name', 'Sex', 'Ticket', 'Embarked', 'Cabin_Serial', 'Cabin']

Salutation Frequencies:

Salutation

Mr 537

Miss 185

Mrs 129

Master 40

Name: count, dtype: int64

Frequency Encoded Columns:

	Salutation	Salutation_Dict	Sex	Sex_Dict	Embarked	Embarked_Dict	Cabin_Serial	Cabin_Serial_Dict	Cabin	Cabin_Dict
0	Mr	537	male	577	S	646	C	746	B96 B98	691
1	Mrs	129	female	314	C	168	C	746	C85	1
2	Miss	185	female	314	S	646	C	746	B96 B98	691
3	Mrs	129	female	314	S	646	C	746	C123	2
4	Mr	537	male	577	S	646	C	746	B96 B98	691

	PassengerId	Survived	Name	Age	SibSp
0	1	0	Braund, Mr. Owen Harris	22.0	1
1	2	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38.0	1
2	3	1	Heikkinen, Miss. Laina	26.0	0
3	4	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35.0	1
4	5	0	Allen, Mr. William Henry	35.0	0

Parch	...	Ord_Pclass	Salutation_Dict	Sex_Dict	Embarked_Dict	Cabin_Serial_Dict	Cabin_Dict
0	...	3	537	577	646	746	691
0	...	1	129	314	168	746	1
0	...	3	185	314	646	746	691
0	...	1	129	314	646	746	2
0	...	3	537	577	646	746	691

Mean Survived per Cabin:

Cabin

A 0.466667

B 0.328338

C 0.593220

D 0.757576

E 0.750000

F 0.615385

G 0.500000

T 0.000000

Name: Survived, dtype: float64

Overall Mean Survived: 0.3838383838383838

```
Encoded Labels Ranks: {'T': 0, 'B': 1, 'A': 2, 'G': 3, 'C': 4, 'F': 5, 'E': 6, 'D': 7}
Target Encoding Result:
  Cabin  Cabin_Encoded
0     B              1
1     C              4
2     B              1
3     C              4
4     B              1
5     B              1
6     E              6
7     B              1
8     B              1
9     B              1
Mean Encoding Result:
     Cabin  Cabin_Mean_Encoded
0  B96 B98            0.303907
1      C85            1.000000
2  B96 B98            0.303907
3     C123            0.500000
4  B96 B98            0.303907
5  B96 B98            0.303907
6      E46            0.000000
7  B96 B98            0.303907
8  B96 B98            0.303907
9  B96 B98            0.303907
Probability Survived / Died Dataframe:
       Survived      Died  Prob_Ratio
Cabin
A      0.466667  0.533333    0.875000
B      0.328338  0.671662    0.488844
C      0.593220  0.406780    1.458333
D      0.757576  0.242424    3.125000
E      0.750000  0.250000    3.000000
F      0.615385  0.384615    1.600000
```

```
G      0.500000  0.500000    1.000000

T      0.000000  1.000000    0.000000

Probability Ratio Dictionary: {'A': 0.875, 'B': 0.488843813387424, 'C':
1.4583333333333333, 'D': 3.125, 'E': 3.0, 'F': 1.6000000000000003, 'G': 1.0,
'T': 0.0}

Probability Ratio Encoding Result:

  Cabin  Encode_Prob_Ratio

0     B           0.488844

1     C           1.458333

2     B           0.488844

3     C           1.458333

4     B           0.488844

5     B           0.488844

6     E           3.000000

7     B           0.488844

8     B           0.488844

9     B           0.488844
```

	PassengerId	Survived	Salutation	Name	Sex	Age
0	1	0	Mr	Braund, Mr. Owen Harris	male	22.0
1	2	1	Mrs	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
2	3	1	Miss	Heikkinen, Miss. Laina	female	26.0
3	4	1	Mrs	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
4	5	0	Mr	Allen, Mr. William Henry	male	35.0
5	6	0	Mr	Moran, Mr. James	male	28.0
6	7	0	Mr	McCarthy, Mr. Timothy J	male	54.0
7	8	0	Master	Palsson, Master. Gosta Leonard	male	2.0
8	9	1	Mrs	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0
9	10	1	Mrs	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0

...	Ticket	Fare	Embarked	Cabin_Serial	Ord_Pclass	Encode_Prob_Ratio
...	A/5 21171	7.2500	S	C	3	0.488844
...	PC 17599	71.2833	C	C	1	1.458333
...	STON/O2. 3101282	7.9250	S	C	3	0.488844
...	113803	53.1000	S	C	1	1.458333
...	373450	8.0500	S	C	3	0.488844
...	330877	8.4583	Q	C	3	0.488844
...	17463	51.8625	S	E	1	3.000000
...	349909	21.0750	S	C	3	0.488844
...	347742	11.1333	S	C	3	0.488844
...	237736	30.0708	C	C	2	0.488844